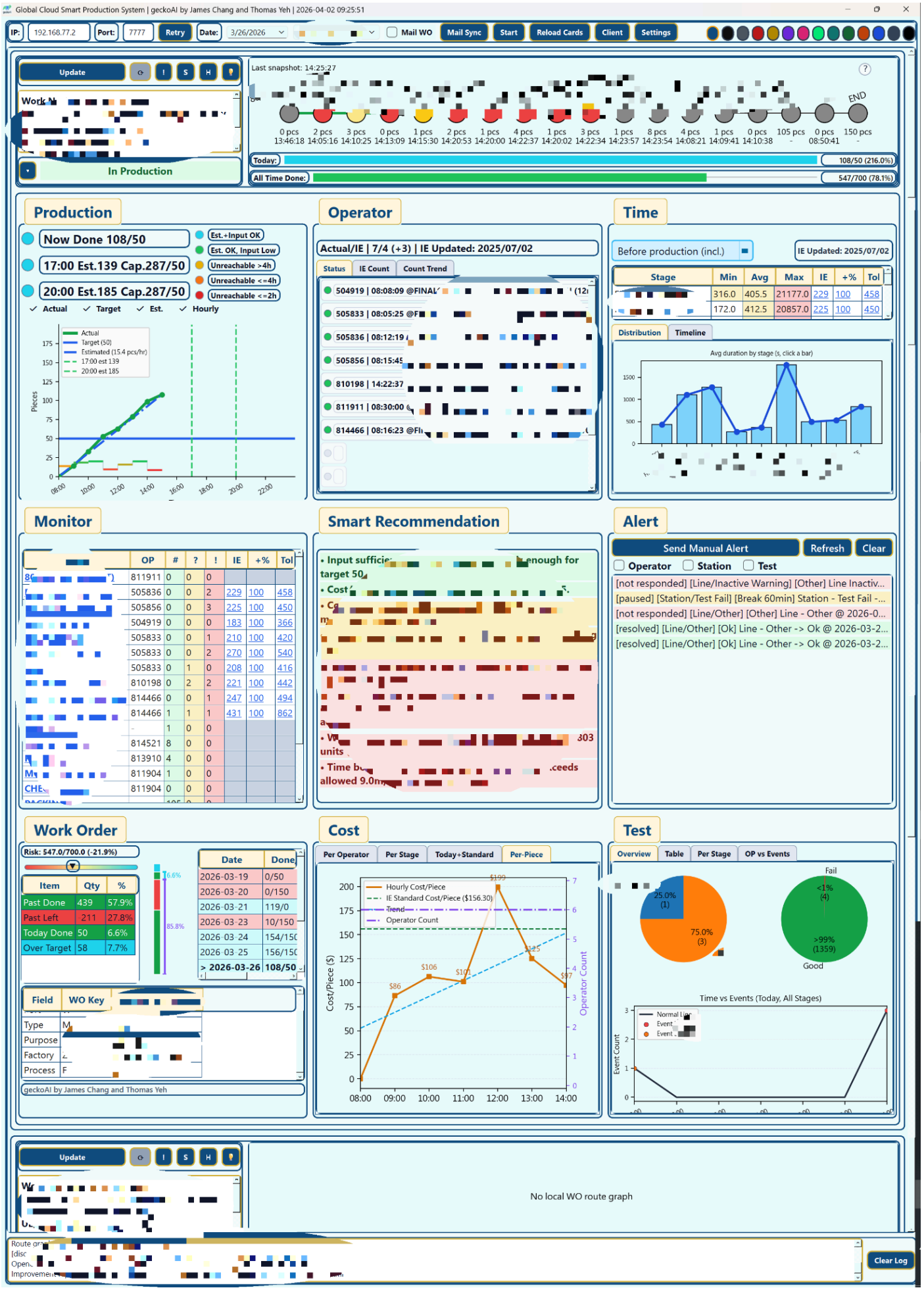
Global Smart Production Monitoring System - User Manual

by James Chang & Thomas Yeh

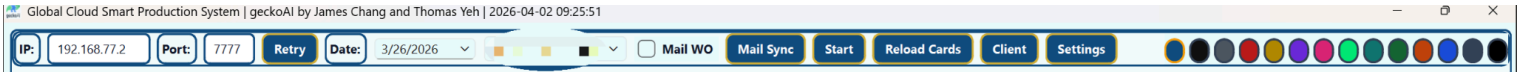
4/2/2026

APP Overview



APP Control

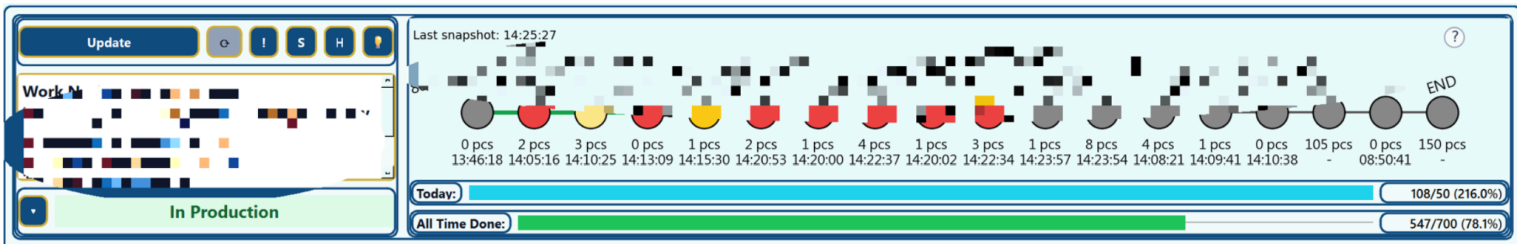
APP Control Bar



IP [textfield] : The destination server IP for data fetching.
Port [textfield] : The destination server port for data fetching.
Retry [button] : Reconnect to the destination server.
Date [date] : The date the app operations reference.
Model [dropdown] : The filter applied to the query based on the model.
Mail WO [tickbox] : The filter applied to the query based on the hourly mail report.
Mail Sync [button] : Manual request to fetch the hourly mail report.
Start [button] : Start the query.
Reload Cards [button] : Rebuild all cards.
Client [button] : Run the "Client" app.
Settings [button] : Open the settings file.
Style Options [button] : Change the appearance of the app.

Work Order Card

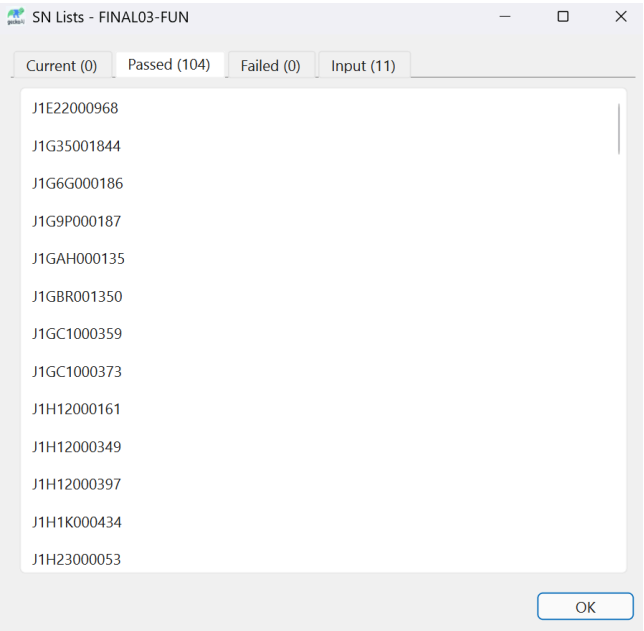
Overview



Update [button] : Updates the data of this work order (non-historical).
Auto Update [button] : Popup, toggle auto-update with a custom interval.
Debug Info [button] : Shows the raw data for this work order.
Settings [button] : Opens up the work-order-specific setting.
Daily Export [button] : Generate and export a daily report.
All-time Export [button] : Generate and export an all-time report.
Prod Station [click] : Right-click on a station to set it to the production station.

Shows the information such as work order details (number, part number, scheduled, and status…etc), station status (hover to see more details), today's progress, and all-time progress.

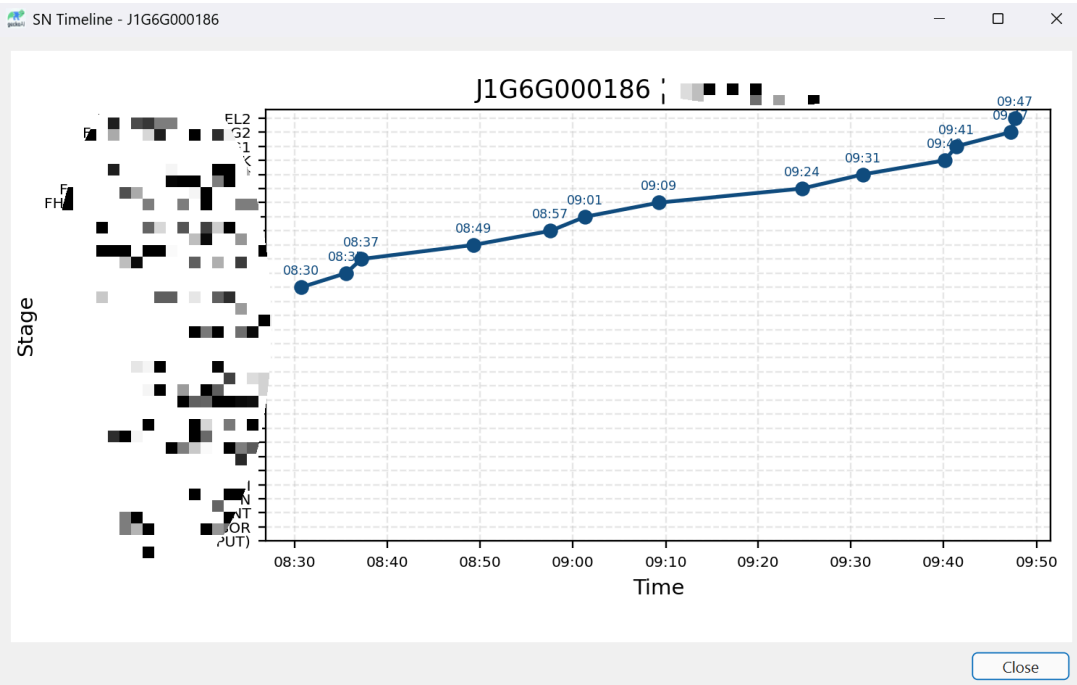
Station SN History



Current : The current DUTs processing or idling at the station.
Passed : The DUTs passed the station.
Failed : The DUTs failed at the station.
Input : The DUTs entered the line from this station today.

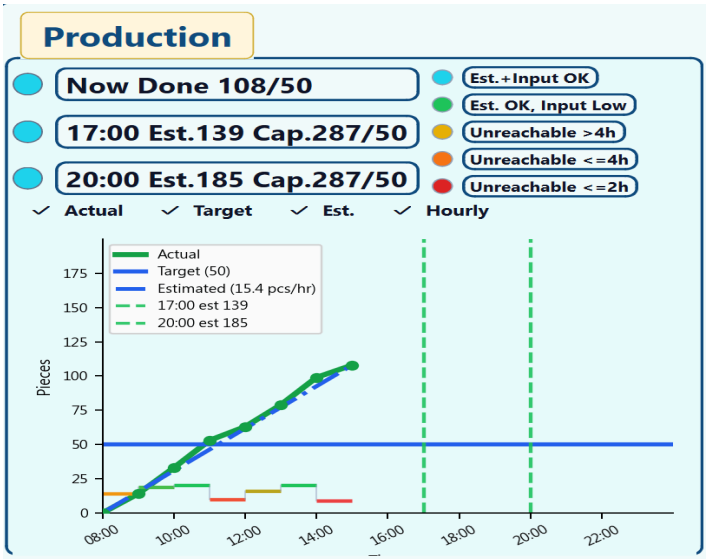
Click on an SN number to view its history.

SN History



View the operating history of the specific SN.

Card Detail - Production

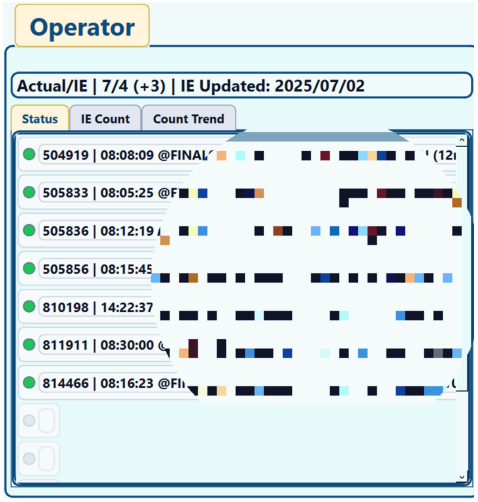


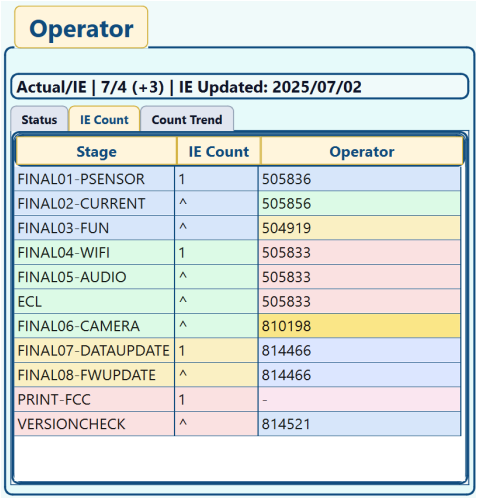
Shows real-time production progress versus plan, with deadline risk indicators (Now, 17:00, 20:00) and trend visualisation.

Functionalities:

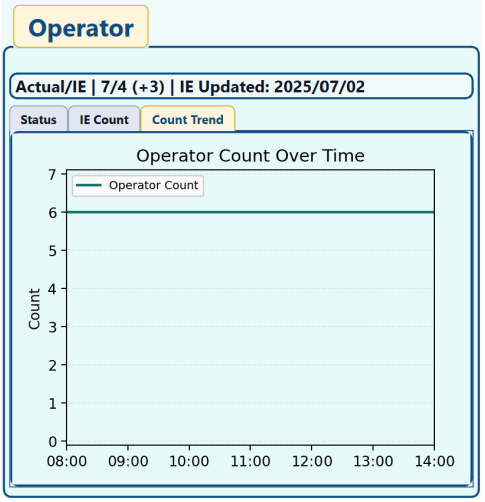
- Displays production trend chart with Actual, Target, Estimated, and Hourly views.
- Uses traffic-light style status for milestone risk at key times.
- Allows users to toggle line layers on/off for focused analysis.
- Supports hover-based insight on trend points.

Card Detail - Operator





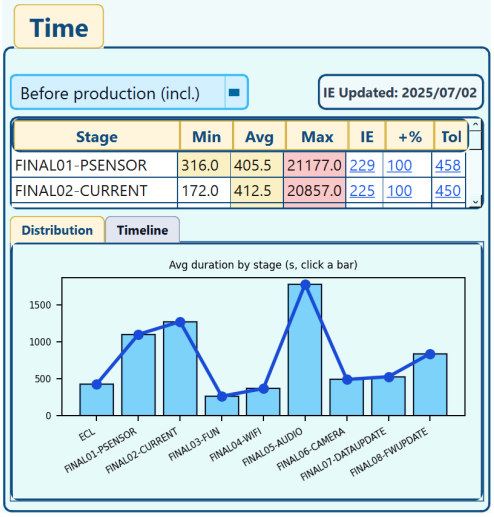
Stage	IE Count	Operator
FINAL01-PSENSOR	1	505836
FINAL02-CURRENT	^	505856
FINAL03-FUN	^	504919
FINAL04-WIFI	1	505833
FINAL05-AUDIO	^	505833
ECL	^	505833
FINAL06-CAMERA	^	810198
FINAL07-DATAUPDATE	1	814466
FINAL08-FWUPDATE	^	814466
PRINT-FCC	1	-
VERSIONCHECK	^	814521

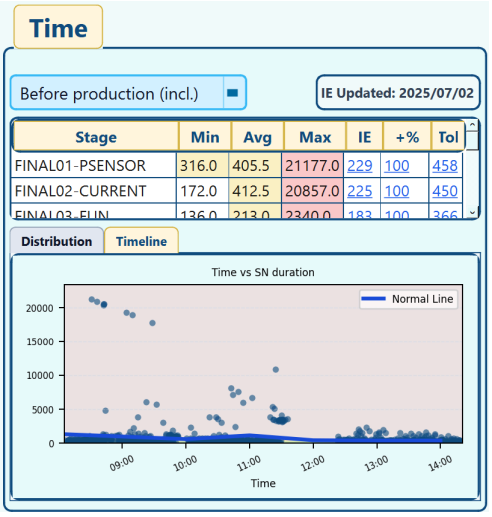


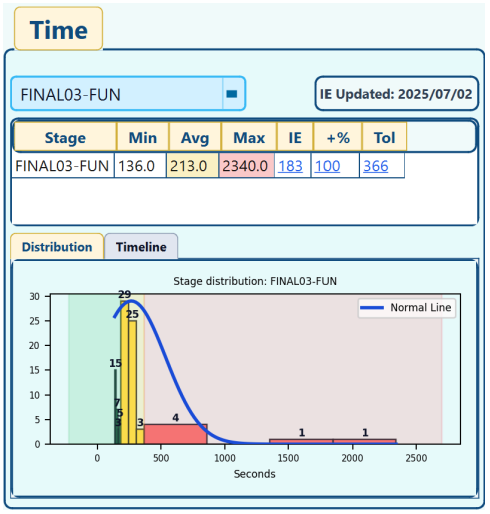
Shows how many people are working on the job over time to help with workforce optimising.

- Shows individual operator status (Active/Idle/Away), and their starting time & station to their last activity time & station
- Shows the standard workforce distribution vs actual.
- Shows the change of actual operator allocation over time.

Card Detail - Time



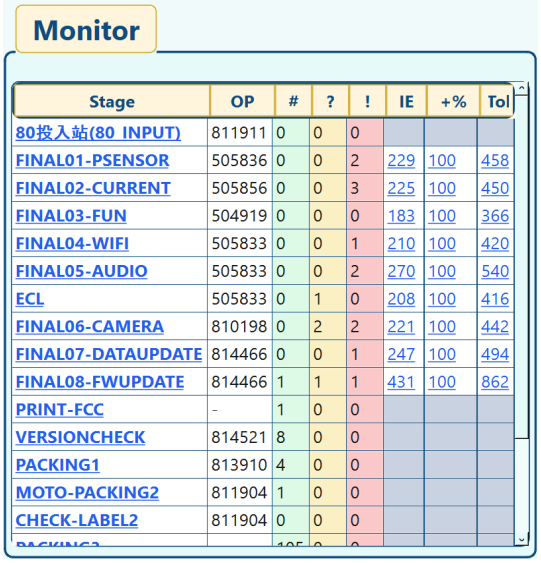


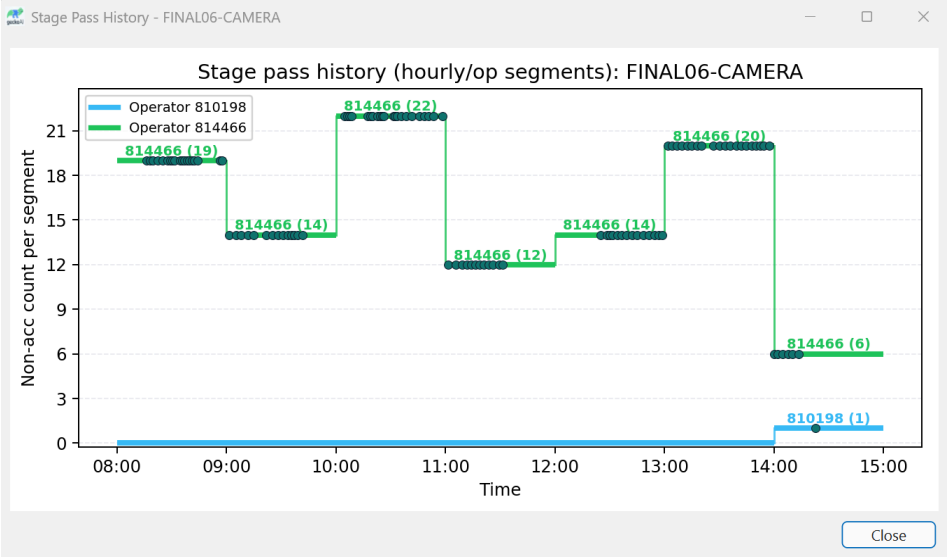


Timing and duration information of overall and individual stations.

- Visualise station operating time distribution.
- Visualise each DUT operating time and timestamp.
- Compare the actual time value against the standard time.

Card Detail - Monitor

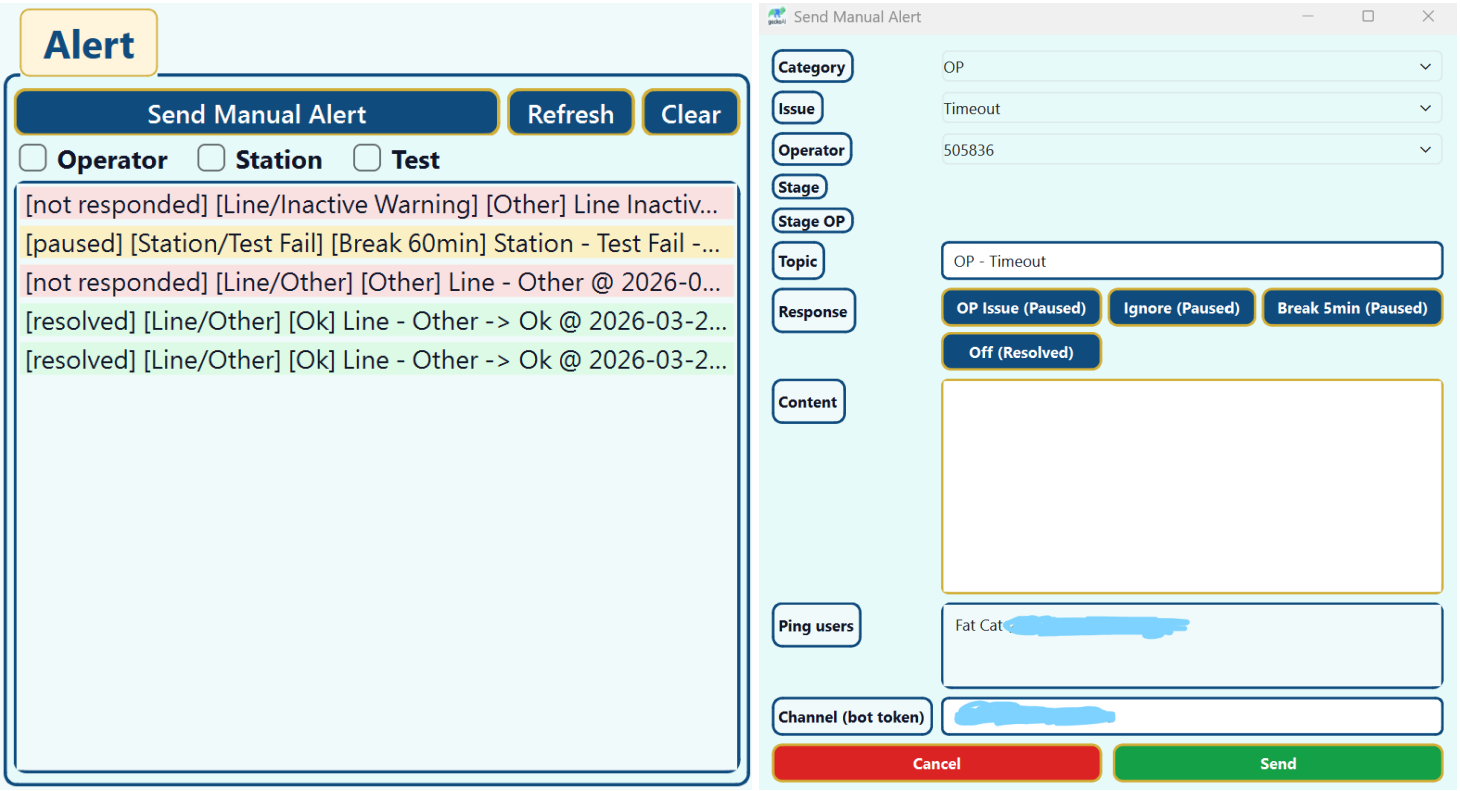




A snapshot of stations' live status.

- View stations' live status against the standards.
- View the individual operating history of productivity and the operator.

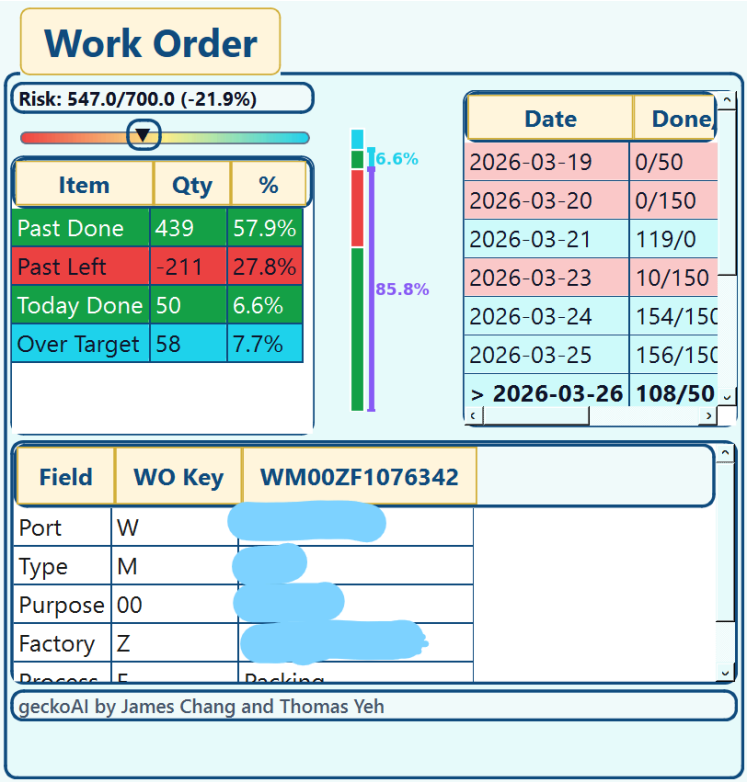
Card Detail - Monitor



Create and manage notifications to draw attention to issues, request help, and look for responses.

- Automated alert system.
- Send an alert manually.
- View alert history and track responses.
- Toggling automated alert for different categories.

Card Detail - Work Order



Core WO details, overall progress, and historical production records.

- View production risks and segments.

- View the done-by-date vs target breakdown.

Card Detail - Cost

Cost

Per Operator | Per Stage | Today+Standard | Per-Piece

OP	Rate ($)	Now ($)	17:00 ($)	20:00 ($)	Target
504919	$315	$1914	$2734	$3679	$12191
505833	$315	$1987	$2807	$3752	$12264
505836	$315	$1858	$2678	$3622	$12135
505856	$315	$1629	$1629	$1629	$1629
810198	$253	$1435	$1435	$1435	$1435
811911	$253	$1488	$2146	$2905	$9742
814466	$253	$1582	$2240	$2999	$9836
TOTAL		$11894	$15668	$20021	$59232

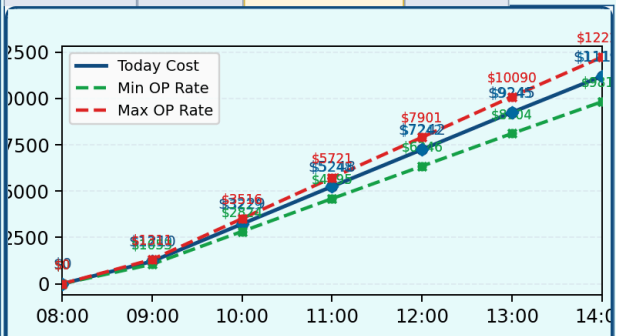
Cost

Per Operator | Per Stage | Today+Standard | Per-Piece

Stage	Trials	Avg (s)	MP	/Trial ($)
FINAL01-PSENSOR	89	405.5	0.866	$25
FINAL02-CURRENT	93	412.5	0.781	$23
FINAL03-FUN	104	213.0	0.362	$5
FINAL04-WIFI	106	117.5	1.526	$13
FINAL05-AUDIO	106	807.5	0.104	$6
ECL	107	334.0	0.237	$6
FINAL06-CAMERA	111	200.0	1.268	$18
FINAL07-DATAUPDATE	110	422.0	0.162	$5
FINAL08-FWUPDATE	108	592.5	0.120	$5

Cost

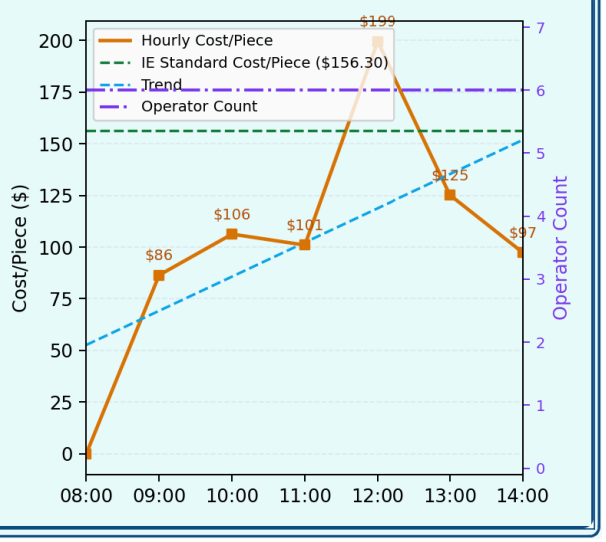
Per Operator | Per Stage | Today+Standard | Per-Piece



Scenario	$253.00	$315.00	Highest	Diff ($)	Diff (%
Max	0	7	$12224	+1031	+9.2%
Actual	3	4	$11192		
Min	7	0	$9818	-1375	-12.3%

Cost

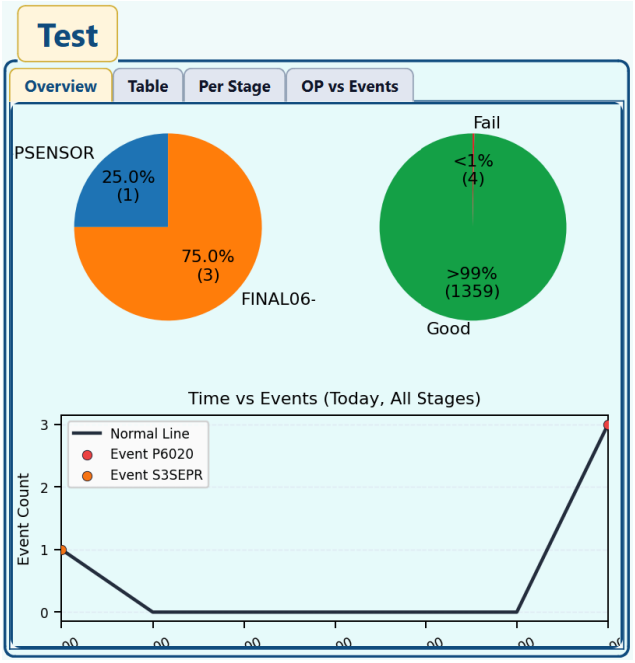
Per Operator | Per Stage | Today+Standard | Per-Piece

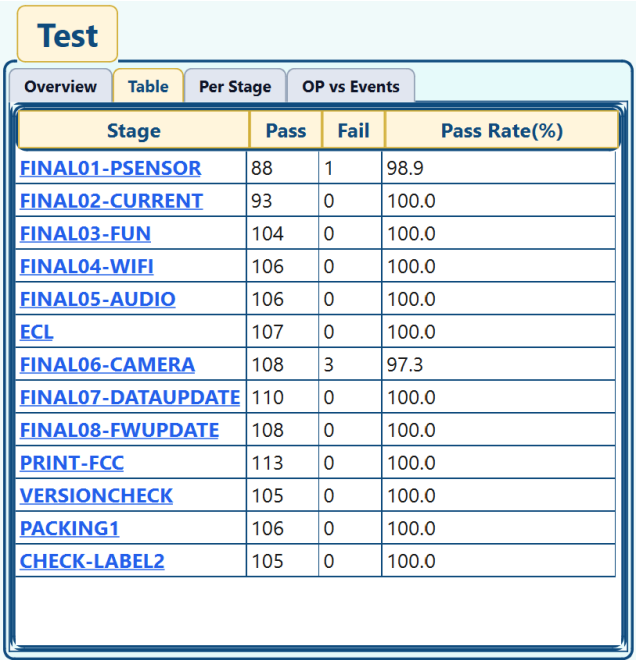


Cost breakdown for each DUT, operator, and station.

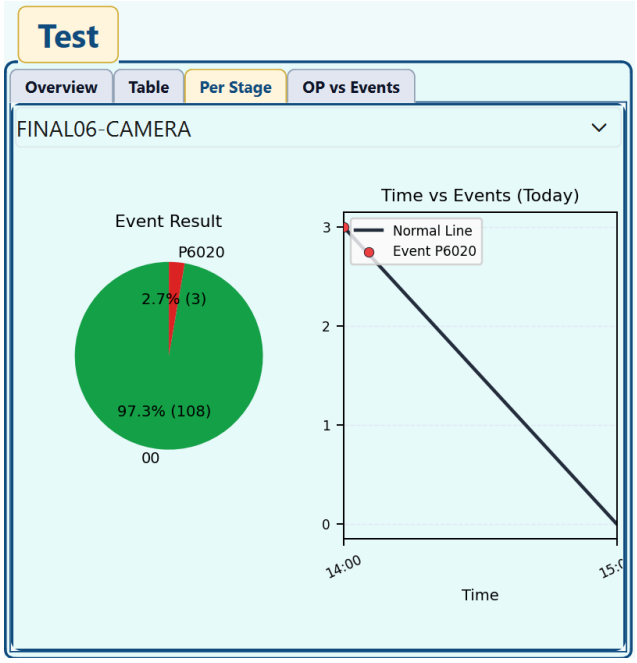
- Provides the estimated operator cost at each deadline.
- Provides the estimated cost per station trial.
- Provides the estimated overall cost for today.
- Provides the estimated cost per DUT each hour today compared to the standard cost.

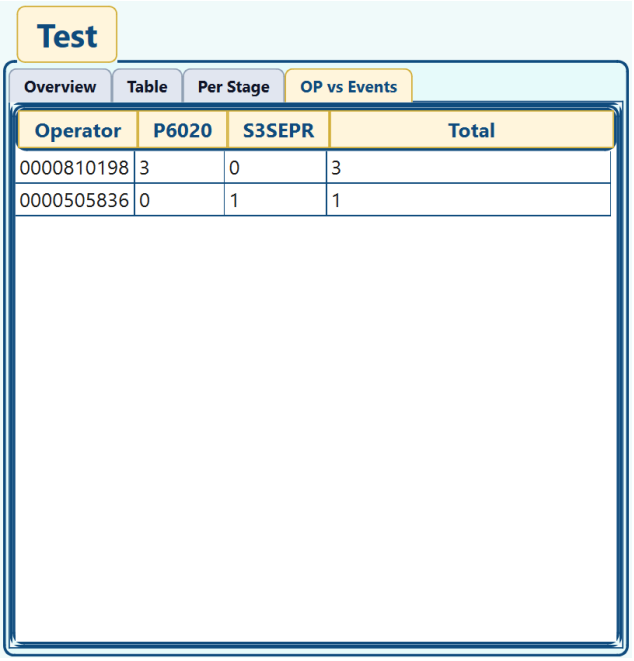
Card Detail - Test





Stage	Pass	Fail	Pass Rate(%)
FINAL01-PSENSOR	88	1	98.9
FINAL02-CURRENT	93	0	100.0
FINAL03-FUN	104	0	100.0
FINAL04-WIFI	106	0	100.0
FINAL05-AUDIO	106	0	100.0
ECL	107	0	100.0
FINAL06-CAMERA	108	3	97.3
FINAL07-DATAUPDATE	110	0	100.0
FINAL08-FWUPDATE	108	0	100.0
PRINT-FCC	113	0	100.0
VERSIONCHECK	105	0	100.0
PACKING1	106	0	100.0
CHECK-LABEL2	105	0	100.0





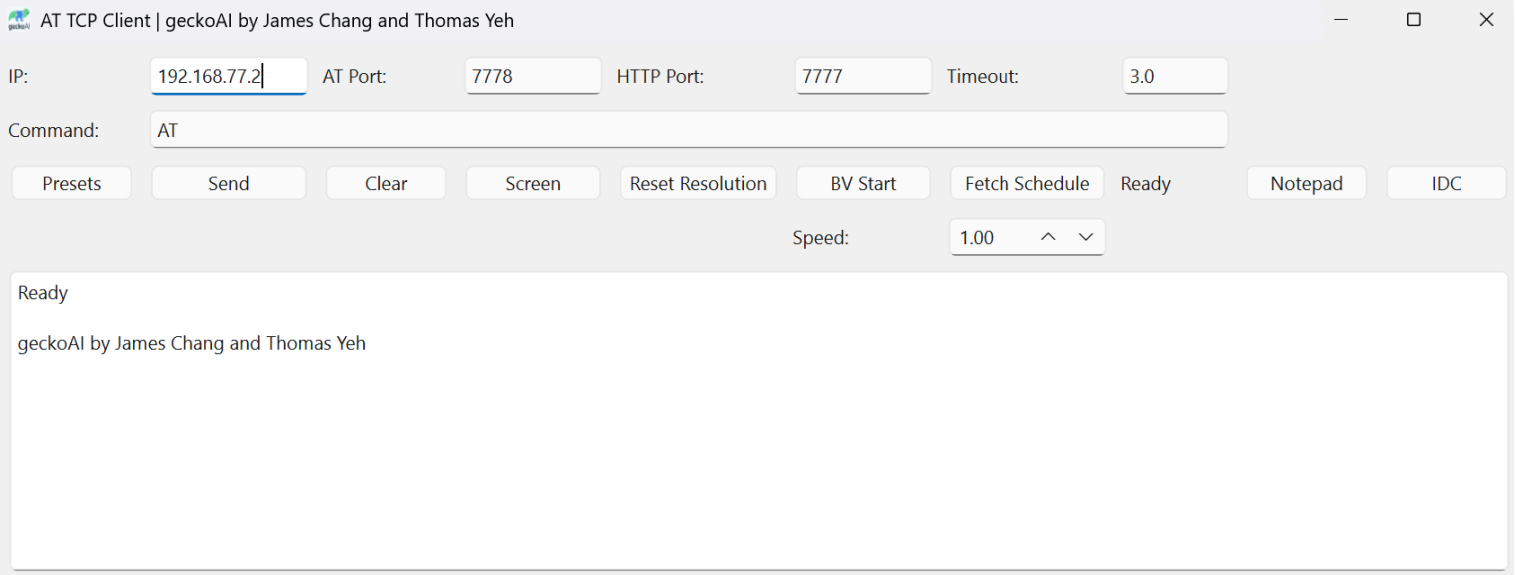
Operator	P6020	S3SEPR	Total
0000810198	3	0	3
0000505836	0	1	1

Test results breakdown for overall and each station.

- View the overall failing station and timeline.
- View the pass rate of each station.
- View the error code breakdown, the operator, and timeline for each station.

Client APP

Introduction



This application is a desktop operations platform for remote data access and AT command automation.
It is designed to help teams operate and monitor existing systems without deep manual intervention, reducing the need for direct hooks into production environments and lowering the risk of unnecessary system exposure.
Instead of relying on ad-hoc remote sessions or repeated manual command input, the app provides a controlled interface to:

- Connect to devices over AT TCP
- Run repeatable automation scripts
- Collect screenshots and operational data
- Sync schedule and mail-derived work-order information
- Generate reports and status outputs for operational follow-up

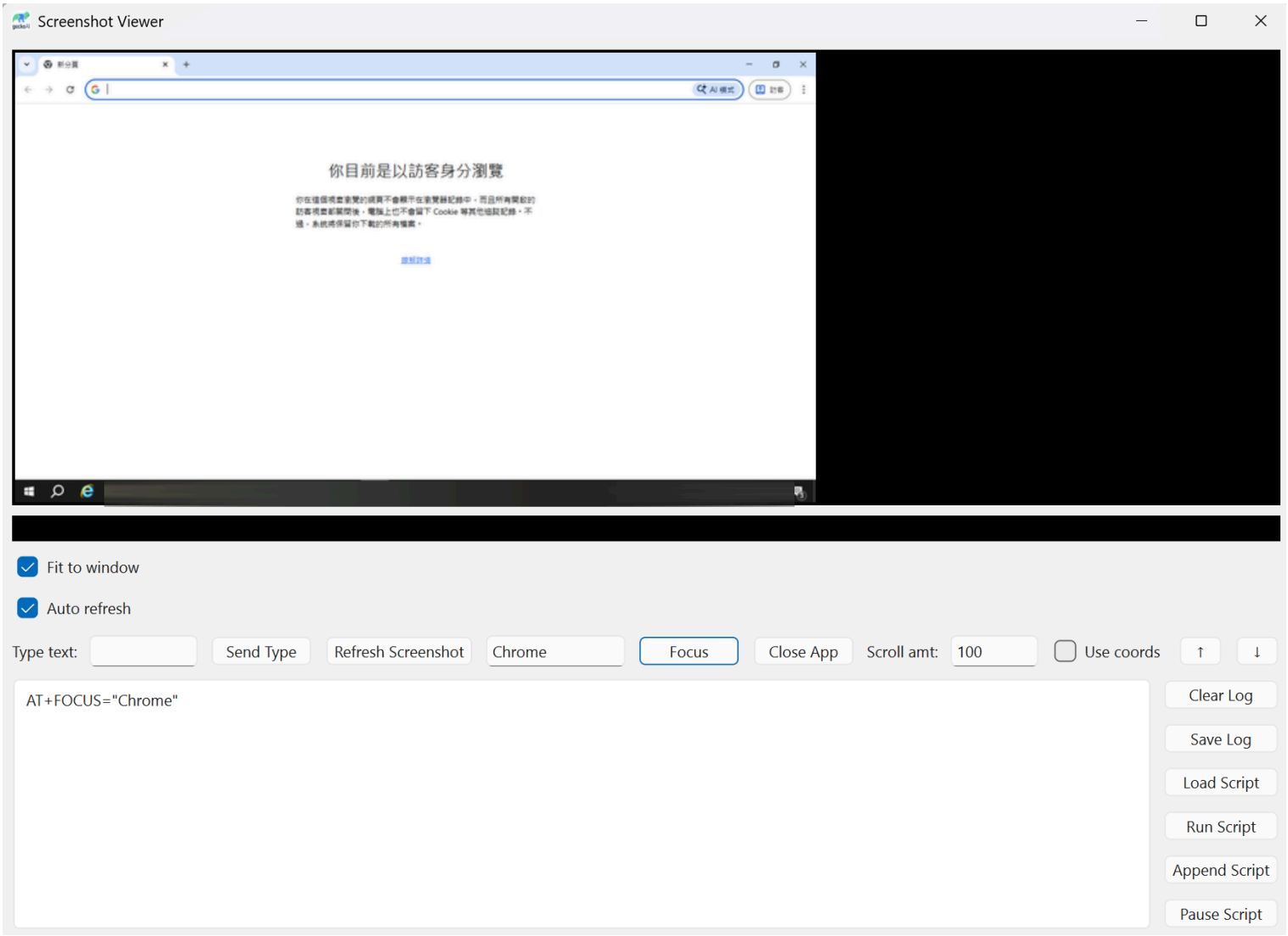
The result is a safer and more standardised workflow: less manual handling, fewer touchpoints on live systems, and more consistent execution of repetitive tasks.

Core Functionalities

AT Command Console

- Send AT commands to remote targets over TCP.
- Configure IP, AT port, HTTP port, and timeout.
- View command responses in real time.

Screenshot-Based Remote Interaction



- Capture and view remote screenshots.
- Perform type, click, drag, type, and scroll actions on screenshot coordinates.
- Optional auto-refresh for near-live operation support.

Command Presets and Script Automation

- Run predefined workflows (for example, WO query flows).
- Execute script files with WAIT/SLEEP control and speed scaling.
- Pause/resume script execution and reuse script logs.
- Variable-based script substitution for reusable automation templates.
- Centralized logs for actions and responses.
- Input/action history for repeatability.
- Structured UI workflows to reduce operator error in repetitive tasks.

Contact

geckoai.contact@gmail.com